

TRANSPORT PHENOMENA I, CHE 311

MIDTERM EXAM 1

CLOSED BOOK, TIME: 75 MINUTES

A liquid film of constant thickness δ is flowing down an inclined flat conveyor belt of length L and width W that is moving with a constant velocity v_0 .

1- Develop a shell momentum balance to obtain the steady-state velocity profile in the liquid film. Assume L and W to be large compared to δ , so the disturbances at the edges of the system may be ignored.

2- Determine the viscous force of the liquid film on a unit area of the conveyor belt.

3- Determine the mass flow rate of the liquid film. $\dot{m} = \rho Q = \rho A \bar{v}$

Your answers may be in terms of the following constant parameters:

μ and ρ : viscosity and density of the liquid

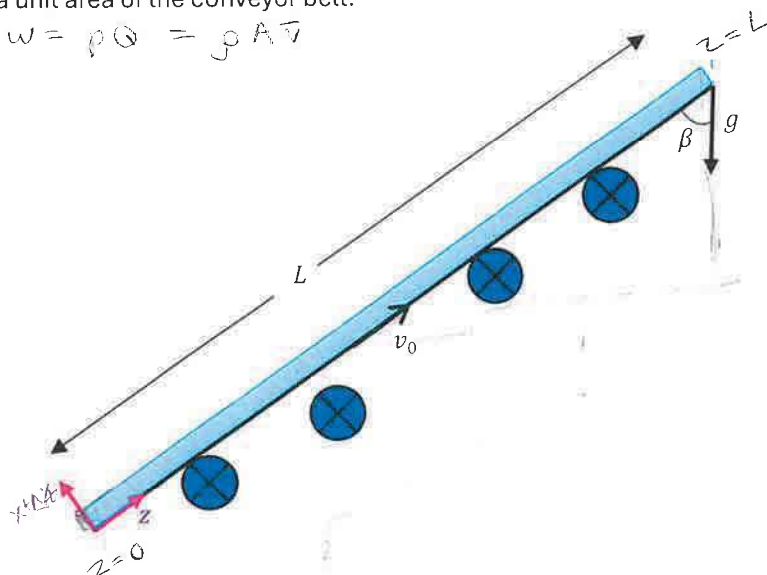
δ : thickness of the liquid film

L and W : length and width of the conveyor belt surface.

v_0 : velocity of the conveyor belt surface in z -direction.

g : gravitational constant.

β : angle between the conveyor belt surface and the direction of gravity.



$$V_x = 0$$

$$V_y = 0$$

$$\rho v_z v_z = \text{constant}$$

at $z=0$ and $z=L$

$$\int_V dV \frac{\partial \rho \langle v \rangle}{\partial t} = - \int_S dS \langle \phi | n \rangle + \int_V dV \rho \langle g \rangle$$

$$|\phi| = \rho \langle v \rangle \langle v \rangle + |\pi| = \rho \langle v \rangle \langle v \rangle + |\tau| + p |\delta|$$

$$|\tau| = -\mu(|\nabla v| + |\nabla v|^T)$$

$$\rho v_i v_j$$

$$\Delta_{\text{convective}} + \Delta_{\text{molecular}} + \text{gravity} = 0$$

$$LW [\tau_{zx}|_x - \tau_{zx}|_{x+\Delta x}] + \Delta x W [p|_{z=0} - p|_{z=L}] +$$

$$LW \Delta x \rho g \cos \beta = 0$$

$$LW \Delta x$$

$$\lim_{\Delta x \rightarrow 0} \left(\frac{\tau_{zx}|_x - \tau_{zx}|_{x+\Delta x}}{\Delta x} + \frac{p|_{z=0} - p|_{z=L}}{L} \right) = -\rho g \cos \beta$$

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$$\frac{\tau_{zx}|_{x+\Delta x} - \tau_{zx}|_x}{\Delta x} - \frac{P|_{z=0} - P|_{z=L}}{L} = \rho g \cos \beta$$

$$\frac{d\tau_{zx}}{dx} - \frac{P|_{z=0} - P|_{z=L}}{L} = \rho g \cos \beta$$

$$\int \frac{d\tau_{zx}}{dx} = \int \rho g \cos \beta$$

$$\tau_{zx} = \rho g \cos \beta x + C_1$$

At $x=0$: $\tau_{zx} = 0$ X

$$C_1 = 0$$

$$\tau_{zx} = -\mu \frac{dv_z}{dx}$$

$$\int \frac{dv_z}{dx} = \int \frac{\rho g \cos \beta x}{\mu}$$

$$v_z = \frac{\rho g \cos \beta x^2}{2\mu} - \frac{\rho g \cos \beta \delta^2}{2\mu}$$

$$1. \quad v_z = \frac{\rho g \cos \beta}{2\mu} (x^2 - \delta^2)$$

$$v_z = -\frac{\rho g \cos \beta x^2}{2\mu} + C^2$$

$x = \delta$ at $v_z = 0$ X

$$0 = -\frac{\rho g \cos \beta \delta^2}{2\mu} + C^2$$

$$C^2 = \frac{\rho g \cos \beta \delta^2}{2\mu}$$

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$$2. \quad v_z = \frac{\rho g \cos \beta}{2\mu} (x^2 - \delta^2)$$

$$\frac{dv_z}{dx} = \frac{\rho g \cos \beta}{\cancel{2\mu}}$$

$$\tau_{zx} = -\mu \frac{dv_z}{dx} \quad \tau_{zx} = -\rho g \cos \beta x = \boxed{-\rho g \cos \beta L}$$

$$3. \quad \text{avg velocity} = \frac{1}{A} \int_A v_z dA$$

$$= \frac{1}{\delta} \int_0^\delta \frac{\rho g \cos \beta x^2}{2\mu} \left(1 - \frac{\delta^2}{x^2}\right) dx$$

$$= \frac{\rho g \cos \beta}{2\mu \delta} \int_0^\delta x^2 - \delta^2 dx = \left[\frac{x^3}{3} - \delta^2 x \right]_0^\delta = \frac{\delta^3}{3} - \frac{\delta^3}{1} = -\frac{2\delta^3}{3}$$

$$= \frac{\rho g \cos \beta}{\cancel{2\mu} \delta} \cdot \frac{-2\delta^3}{3} = \frac{\rho g \cos \beta \delta^2}{3\mu}$$

$$A = \delta W$$

$$W = \rho \cdot A \cdot \bar{v} = \boxed{\frac{\rho^2 g \cos \beta \delta^3 W}{3\mu}}$$